

Characterization of eight polymorphic loci for Maasai giraffe (*Giraffa camelopardalis tippelskirchi*) using non-invasive genetic samples

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Abstract We used Illumina sequencing to develop eight novel microsatellite loci for Maasai giraffe (*Giraffa camelopardalis tippelskirchi*), and screened them using fecal DNA. Genetic diversity was assessed for 40 individuals from the Katavi National Park/Rukwa Game Reserve ecosystem in Tanzania. The number of alleles per locus ranged from 2 to 14 (mean = 6.2) and mean expected heterozygosity was moderate (range 0.025–0.838, mean = 0.527). These markers were successfully tested using degraded DNA and may be useful for future studies of giraffe, especially those that use non-invasive sampling techniques.

Keywords *Giraffa camelopardalis* · Microsatellite · Illumina sequencing · Non-invasive sampling

The IUCN classifies giraffe as a species of least concern due to their wide geographic range and estimate of 80–100,000 individuals worldwide (IUCN 2011). However, multiple subspecies have been characterized based on pelage characteristics and genetic analyses (Dagg and Foster 1976; Kingdon 1997; Brown et al. 2007), and the IUCN does not assess all of these subspecies individually. Although some

subspecies appear to be stable, others are declining due to habitat loss or poaching (Fennessy and Brown 2010). The Maasai subspecies (*Giraffa camelopardalis tippelskirchi*) is found throughout Tanzania and southern Kenya, and while assumed to be widely distributed within protected areas, no recent population estimates have been gathered. Conservation of this subspecies would be aided by knowledge of the genetic diversity and structure within populations. We used Illumina sequencing to design 8 novel microsatellite loci for Maasai giraffe and characterized them using noninvasive samples. Previous studies have presented microsatellite primers for Maasai giraffe (Huebinger et al. 2002; Brown et al. 2007), but several of those loci did not amplify in Maasai giraffe from southern Tanzania, while others were monomorphic. Carter et al. (2012) developed markers for Angolan giraffe (*G. c. angolensis*) and screened them using DNA extracted from tissue. This study presents the first suite of giraffe primers verified to amplify from relatively degraded DNA.

Genomic DNA was extracted from a tissue sample collected from a Maasai giraffe carcass in the Ruaha ecosystem, Tanzania, using a DNEasy blood and tissue kit (Qiagen). For Illumina library preparation, genomic DNA was sheared to a median size of 300–400 bp with a Bio-Ruptor sonicator (Diagenode Inc.). Sheared DNA was converted to Illumina libraries as previously described (Cronn et al. 2008). Library construction followed the recommended Illumina protocols, with the exception that index sequencing (“barcoded”; Cronn et al. 2008; Craig et al. 2008) adapters were substituted for standard paired-end adapters. Indexed libraries were pooled in approximately equimolar concentrations and submitted for paired-end 80 bp sequencing. Cluster generation and sequencing used Illumina version 3.0 reagents, and image acquisition and base calling used the Illumina pipeline version 1.5.

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We designed primers for 36 loci with Primer 3 (Rozen and Skaletsky 2000) and used fluorescein-labelled 12-dCTP (Perkin Elmer) in a PCR to test for polymorphism across 8 samples. Amplification was performed in 15 µL reactions consisting of 1× Magnesium-free buffer, 2.25 mM MgCl₂, 160 µM of each dNTP, 7.5 µg bovine serum albumin, 0.1 µM of each primer, 0.7U Apex Hot Start Taq polymerase, 0.5 nmol d-CTP, and approximately 50 ng of genomic DNA. Thermal cycling conditions included: initial denaturation of 15 min at 95 °C, followed by 40 cycles of 95 °C for 30 s, 45 s at a locus-specific temperature (Table 1), and extension at 72 °C for 30 s, with final elongation of 5 min at 72 °C. We analyzed samples on an ABI3730 DNA analyser with LIZ500 sizing standard and verified fragment sizes using Genemapper 4.1 (Applied Biosystems).

Eight of the 36 primers were polymorphic, scoreable, and amplified a single product. We characterized these loci with fecal DNA samples from 40 individuals from the Katavi National Park/Rukwa Game Reserve ecosystem, Tanzania. We extracted DNA from fecal samples using an AquaGenomic extraction kit (Multitarget Pharmaceuticals). For each locus, one primer was fluorescently tagged with NED, PET, VIC (Applied Biosystems) or 6-FAM (Sigma-Aldrich).

We used BLAST to compare sequences from our loci with those published on GenBank and in supplementary material of Carter et al. (2012) to verify that our loci were unique. We used Genepop v.4 (Raymond and Rousset 1995) to test for evidence of linkage disequilibrium between all pairs of loci and for departures from Hardy–Weinberg Equilibrium. We quantified allelic diversity (A), observed heterozygosity (H_O), and expected heterozygosity (H_E) using GENALEX (Peakall and Smouse 2006). There was no evidence of linkage disequilibrium between any loci, nor did any locus exhibit departures from HWE after sequential Bonferroni correction (Table 1). Locus Gica16120 had two alleles, one of which was seen in only one individual and thus the probability of HWE could not be calculated (Table 1). However, samples collected across a larger area of central Tanzania exhibited a third allele at this locus (Crowhurst 2012).

Understanding the genetic patterns of populations can be critical for management programs and methods that incorporate non-invasive sampling techniques are especially important for species that may be difficult to sample directly due to financial or logistical constraints. Having verified the ability of these markers to genotype degraded DNA, we believe that they may aid future studies of giraffe.

Table 1 Characterization of eight novel microsatellite loci in Maasai giraffe (*G. c. tippelskirchi*) from the Katavi National Park/Rukwa Game Reserve ecosystem in Tanzania (n = 40)

Locus	Primer sequence (5′–3′)	Repeat motif	T _A (°C)	No. of alleles	Fragment size (bp)	H _O	H _E	Probability of HWE	GenBank accession no.
Gica13905	F: CAG ACA GAT GGG GAA ACT GAG R: TTT GGC TAA ATT TTT CAT ACA CAC A	(GT) ₇ A(TG) ₁₇	63	3	230–236	0.350	0.363	0.809	JX424290
Gica13619	F: CAG GTT TTC ATT GTA TTG CTC TG R: ATG CAG AAT GGG GGT TAC AG	(TG) ₁₀	60	3	257–261	0.450	0.444	0.856	JX424289
Gica9976	F: GGG AGG AGA CTG GAT TGT CA R: AGT GGC TCT CCA AAG CAC AT	(GT) ₁₈	63	8	250–272	0.700	0.823	0.167	JX424293
Gica16120	F: AAA GTA ATT TGG GCA AAT GTG G R: TTT GGC CAG TCT TCA GAT CA	(TG) ₆ A (GT) ₁₃	60	2	134–140	0.025	0.025	–	JX424295
Gica16160	F: TGC AGA GCA ATT GCA AAC AT R: GTG GGC AAC TGT TCA TAG GG	(GT) ₂₄	60	10	133–167	0.850	0.838	0.474	JX424296
Gica10894	F: TGT TGT CAC TTA CCC GTT TTC C R: AGA GTC TGG GAT GCA TTT GG	(GT) ₁₆	63	5	240–254	0.400	0.436	0.163	JX424294
Gica9905	F: ATG ATA TTC AGC TGG GCC TCT R: CCT GAT GGA CAC CAG GTT G	(TG) ₁₁ A(GT) ₂	60	14	289–327	0.850	0.821	0.811	JX424292
Gica14170	F: GTG AGG TGC CAT CAC CTT CT R: CAC TGG AGG CAA GTC AAC AA	(AT) ₃ (GT) ₁₈	63	5	261–269	0.475	0.469	0.453	JX424291

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